Pothole Detection Using YOLOv3

Abstract

Potholes are a major safety concern for transportation systems, as they cause vehicle damage, increase accident risk, and demand significant repair resources. Deep learning models, particularly YOLO-based detectors, provide fast and effective solutions for real-time pothole detection. This project evaluates YOLOv3 performance across multiple weather and lighting conditions, trained on Pothole600 and a multi-weather dataset, and tested. Through precision, recall, F1-score, PR curves, and confusion matrices, we analyze the model's strengths and weaknesses. The results show that YOLOv3 performs very well in clear daylight but struggles with low illumination, rain reflections, and domain shifts. The goal of this report is to concisely present methodology, training design, dataset usage, and performance insights.

Introduction

Pothole detection is an essential component of intelligent road monitoring systems. Manual road inspection is slow, expensive, and inconsistent. As a result, computer vision approaches—especially those involving deep neural networks—are increasingly adopted for automated detection.

YOLOv3 is appealing for this task because it combines high detection accuracy with real-time inference speed. However, environmental variations such as rain, shadows, and nighttime lighting can significantly affect model performance. This project systematically evaluates YOLOv3 under different operating conditions and examines how well it generalizes to completely unseen data.

YOLOv3 Methodology

Architecture Overview. YOLOv3 uses Darknet-53 as its feature extractor, consisting of 53 convolutional layers with residual connections. These layers capture both low-level and high-level spatial features.

Three detection scales (13×13, 26×26, 52×52) enable the model to detect potholes of various sizes. This is particularly useful because potholes often appear as small, irregular shapes that vary greatly in size depending on the camera distance and image resolution.

Bounding box regression produces:

$$(t_x, t_y, t_w, t_h, \sigma(t_o)),$$

which define object location and confidence. YOLOv3 also uses anchor boxes and non-max suppression to refine predictions.

Training Configuration.

- Image size: 416×416
- Batch size: 32
- Epochs: 100
- Optimizer: SGD with momentum 0.9
- Data augmentation: Mosaic, HSV shift, cropping, horizontal flip
- Pretrained weights: yolov3.pt

These settings help YOLOv3 adapt to variations while maintaining stable convergence.

Evaluation Metrics

To measure performance, the following metrics were used:

- **Precision:** proportion of correct pothole detections.
- **Recall:** ability to find all actual potholes.
- **F1-score:** balance between precision and recall.
- **Confusion Matrix:** visual summary of TP, FP, FN.
- **PR Curve:** shows the trade-off between precision and recall under different confidence thresholds.

Results (Testing)

Only the strongest, most representative results for each condition are included for clarity.

Pothole600

This dataset is clean, well-lit, and consistently labeled, representing an ideal training scenario.

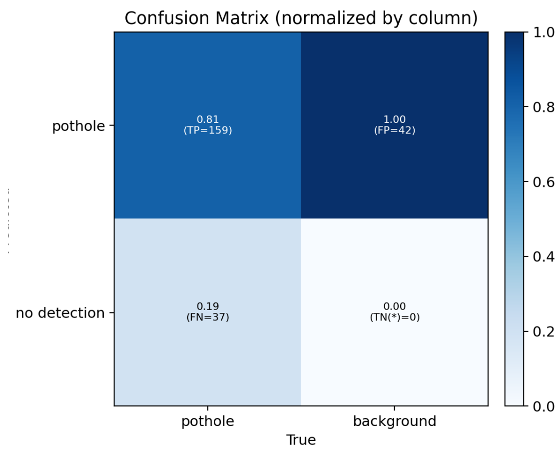

Figure 1: Confusion Matrix for Pothole600

The confusion matrix reflects few false positives or missed detections, confirming YOLOv3's strong baseline performance.

Clear Weather

Under clear lighting, the model performs with its highest consistency. Edges and textures are well-defined, making pothole boundaries easier for YOLOv3 to detect.

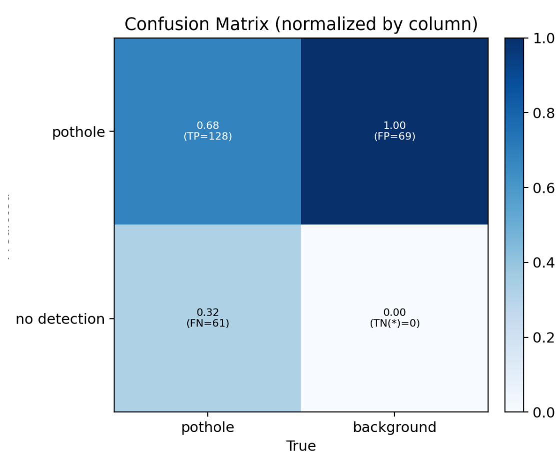

Figure 2: Confusion Matrix for Clear Weather

Rainy Weather

Rain introduces reflections, glare, and surface distortion. These interfere with YOLOv3's ability to correctly identify pothole boundaries, often lowering recall due to missed detections.

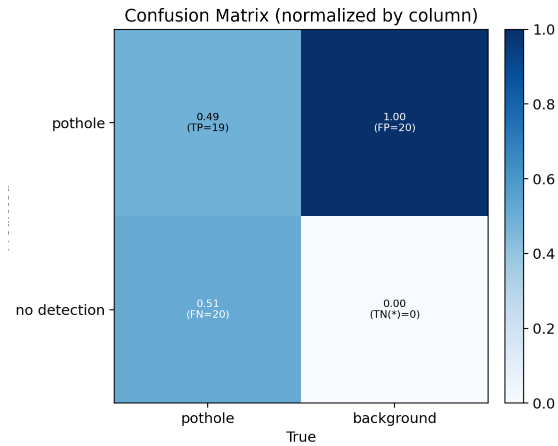

Figure 3: Confusion Matrix for Rainy Conditions

Increased false negatives are observed, showing that wet road surfaces reduce model confidence.

Night Conditions

Low-light images create high contrast and shadow noise. YOLOv3 struggles to generalize in such conditions, leading to a drop in recall and occasional misclassifications.

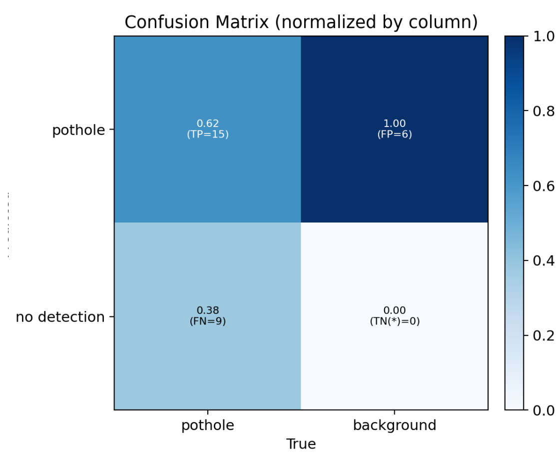

Figure 4: PR Curve for Night Conditions

Potholes become harder to detect as the visible gradient between road and pothole surface decreases.

Sunset Conditions

Sunset lighting produces elongated shadows and uneven illumination, which can cause false detections.

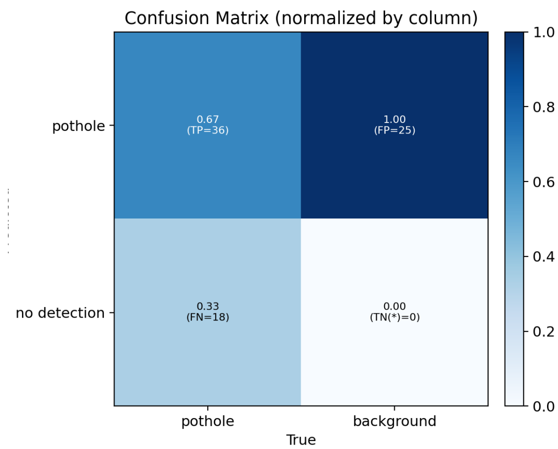

Figure 5: Confusion Matrix for Sunset Conditions

The confusion matrix indicates moderate performance, with occasional misidentification caused by shadow textures.

UNL Real-World Dataset (Robustness Test)

This dataset was never used during training. It contains varied angles, lighting differences, and surface textures not present in the training datasets.

Figure 6: Sample YOLOv3 Predictions on UNL Data

Figure 7: Sample YOLOv3 Predictions on UNL Data

The drop in performance highlights YOLOv3's sensitivity to domain shift. Shadows, new road textures, and unfamiliar camera angles lead to both false positives and missed potholes.

Conclusion

YOLOv3 demonstrates strong accuracy in controlled and clear environments. However, rain, low light, and domain shift significantly reduce the performance. These findings suggest that real-world deployment requires broader training diversity.
Future improvements:

- Expand training with diverse weather and nighttime samples.
- Use domain adaptation or synthetic augmentation.
- Explore newer detection models with stronger robustness.

Additional Sample Results (Appendix)

This section contains additional qualitative samples from each dataset. These examples help visualize how YOLOv3 performs under different lighting and weather conditions.

Pothole600 Samples

Figure 8: Additional detection samples from the Pothole600 dataset.

Clear Weather Samples

Figure 9: Sample detections under clear weather conditions.

Rainy Weather Samples

Figure 10: Detection results from rainy weather images.

Night Conditions Samples

Figure 11: YOLOv3 performance during nighttime scenarios.

Sunset Conditions Samples

Figure 12: Detections captured under sunset lighting conditions.